

Information Communication Technology

Course: CPA — Certified Public Accountant

Level: Foundation Level (CA16)

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Syllabus version: Based on the standard KASNEB CPA Foundation Level Information Communication Technology syllabus topics (CA16) — computer fundamentals, application software (including spreadsheets), computer networks, and information/data security. Authored from general ICT knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

Original work notice. Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

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1. Computer hardware and software

Hardware is the physical, tangible components of a computer (input devices like a keyboard/mouse, output devices like a monitor/printer, the CPU, and storage devices).

Software is the intangible instructions directing hardware — **system software** (e.g. the operating system, managing hardware and providing the platform for other software) and **application software** (programs for specific tasks, e.g. spreadsheets or accounting packages).

2. Data versus information

Data is raw, unprocessed facts with no inherent meaning until organised (e.g. a list of individual transactions). **Information** is data that has been processed, organised and structured to become meaningful for decision-making (e.g. a monthly management report).

Good information should be accurate, timely, relevant, complete, and appropriately presented for its intended user.

3. Application software for accounting and business

- **Spreadsheets** — organising data into rows/columns, with built-in formulas, widely used for budgets and calculations.
 - **Accounting/ERP packages** — recording transactions and generating financial statements, often integrating multiple business functions (finance, inventory, payroll) into one system.
 - **Word processors and presentation software** — documents and reports; slide-based presentations for meetings.
 - **Database software** — storing and retrieving structured data efficiently, useful for larger volumes of customer or transaction records than a spreadsheet can conveniently manage.
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4. Spreadsheets — formulas and functions

A spreadsheet organises data into a grid of rows and columns, each intersection called a **cell**.

Common functions: `=SUM(range)` totals a range of cells; `=AVERAGE(range)` computes the mean; `=IF(condition, value_if_true, value_if_false)` applies conditional logic (e.g. flagging overdue accounts). Spreadsheets are widely used in accounting-support work for budgets, simple ledger summaries, and "what-if" analysis (changing one input to see the effect on a result, e.g. testing how a change in sales volume affects profit).

5. Databases, basics

A **database** is an organised collection of structured data, typically arranged into **tables** (rows and columns), allowing efficient storage, retrieval and querying of large volumes of information — for example, a customer database storing each customer's details and transaction history, queryable to answer specific questions (e.g. "which customers have overdue balances above KSh 50,000?") far more efficiently than searching a large spreadsheet manually.

6. Computer networks and the internet

A **network** connects computers to share data, software and hardware resources. A **LAN** covers a small area (e.g. one office); a **WAN** covers a large area (the internet being the largest WAN). The internet enables business communication, research, online banking and e-commerce; **email** remains one of the most widely used business communication tools, valued for speed and a written record, but requiring care around tone and security (phishing).

7. Information and data security

Key measures: **passwords/access controls** (restricting system access to authorised users), **antivirus/anti-malware software**, **regular backups** (enabling recovery if data is lost or corrupted, including from ransomware), **firewalls** (controlling network traffic), and **user awareness training** (since human error, e.g. falling for phishing or sharing passwords, is a common cause of security breaches alongside technical weaknesses).

8. Accounting information systems, introduction

An **accounting information system (AIS)** is the combination of people, processes, and technology an organisation uses to collect, record, process and report financial data. A well-designed AIS incorporates internal controls (e.g. restricting who can post journal entries, requiring approval for payments above a threshold) directly into the system's design, reducing reliance on manual checking alone and helping ensure the accuracy and integrity of the financial data it produces.

9. Full worked question — spreadsheet formulas for a simple ledger summary

Scenario. A spreadsheet lists four months of expenses in cells D2 to D5: 18,000; 21,500; 16,200; 19,800 (KSh).

Required: (a) Write the formula for cell D6 to compute total expenses. (b) Write the formula for cell D7 to compute average monthly expenses. (c) Compute the resulting values.

Solution:

(a) `=SUM(D2:D5)`

(b) `=AVERAGE(D2:D5)`

(c) Total = 18,000+21,500+16,200+19,800 = **KSh 75,500**. Average = 75,500 ÷ 4 = **KSh 18,875**.

10. Practice examination — KASNEB-style paper (original, not a past paper)

Note on format. This section is laid out the way a KASNEB CPA examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

CPA FOUNDATION LEVEL — INFORMATION COMMUNICATION TECHNOLOGY (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

(a) Distinguish between hardware and software, with two examples of each. (12 marks)

(b) Distinguish between data and information, with one example of each. (8 marks)

(Total: 20 marks)

QUESTION TWO

A spreadsheet lists five weeks of sales in cells E2 to E6: 30,000; 45,000; 28,000; 50,000; 32,000.

(a) Write the formula for cell E7 to compute total sales. (3 marks)

- (b) Write the formula for cell E8 to compute average weekly sales. *(3 marks)*
- (c) Compute the values for E7 and E8. *(6 marks)*
- (d) Outline FOUR common application software packages used in business, with one use of each. *(8 marks)*
- (Total: 20 marks)*
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QUESTION THREE

- (a) Explain what a database is and outline TWO advantages of a database over a spreadsheet for storing large volumes of customer records. *(10 marks)*
- (b) Distinguish between a LAN and a WAN. *(6 marks)*
- (c) Outline TWO risks of using email for business communication. *(4 marks)*
- (Total: 20 marks)*
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QUESTION FOUR

- (a) Outline FOUR measures a business can take to protect its information and data security. *(16 marks)*
- (b) Explain why user awareness training is an important part of information security. *(4 marks)*
- (Total: 20 marks)*
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QUESTION FIVE

- (a) Define an "accounting information system" (AIS). *(6 marks)*
- (b) Explain how internal controls can be incorporated directly into the design of an AIS, with one example. *(8 marks)*
- (c) Outline TWO qualities of good information. *(6 marks)*
- (Total: 20 marks)*
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11. Marking guide — model answers to the practice examination

QUESTION ONE

- (a) *(12 marks)* — Hardware: physical components (e.g. keyboard, monitor). Software: instructions/programs directing hardware (e.g. operating system, spreadsheet program) — 6

marks each for definition plus two examples.

(b) (8 marks) — Data: raw unprocessed facts (e.g. a list of transactions). Information: processed, organised data that is meaningful for decision-making (e.g. a monthly report).

QUESTION TWO

(a) (3 marks) — `=SUM(E2:E6)`

(b) (3 marks) — `=AVERAGE(E2:E6)`

(c) (6 marks) — Total = 30,000+45,000+28,000+50,000+32,000 = **KSh 185,000**. Average = 185,000 ÷ 5 = **KSh 37,000**.

(d) (8 marks — 2 marks each) — Spreadsheet (budgets/calculations); accounting/ERP package (recording transactions); word processor (documents); presentation software (slides for meetings) — database software also acceptable.

QUESTION THREE

(a) (10 marks) — A database is an organised collection of structured data, typically in tables, allowing efficient storage, retrieval and querying; advantages over a spreadsheet: more efficient querying/filtering of large data volumes, and better handling of relationships between different data sets (e.g. customers and their transactions) — any two, 3 marks each, plus 4 for the definition.

(b) (6 marks) — A LAN covers a small geographic area (e.g. one office); a WAN covers a large geographic area (the internet being the largest WAN).

(c) (4 marks — 2 marks each) — Risk of phishing/malicious attachments; risk of miscommunication due to absent tone/non-verbal cues.

QUESTION FOUR

(a) (16 marks — 4 marks each) — Passwords/access controls, antivirus/anti-malware, regular backups, firewalls.

(b) (4 marks) — Because human error (e.g. falling for phishing, sharing passwords) is a common cause of security breaches, so trained, alert users are an important defence alongside technical controls.

QUESTION FIVE

(a) (6 marks) — An AIS is the combination of people, processes and technology an organisation uses to collect, record, process and report financial data.

(b) (8 marks) — Internal controls can be built directly into an AIS's design, e.g. requiring a second-level approval within the system before a payment above a set threshold can be processed, reducing reliance on manual checking alone.

(c) (6 marks — 3 marks each) — Accurate, timely (relevant or complete also acceptable).

12. Exam technique and revision checklist

- Hardware vs software: always pair the definition with a concrete example — a bare definition alone often earns partial marks only.
- Spreadsheet formula questions: write the exact formula syntax, not a description of the calculation in words.
- Data vs information: remember information is data that has been processed/organised to become meaningful — a very commonly tested distinction.
- Security questions: name specific measures (passwords, antivirus, backups, firewalls) rather than a vague "keep it secure."
- AIS questions: always give a concrete example when asked to explain how a control is built into a system, not just an abstract description.

Quick revision — one line per topic:

1. Hardware = physical components; software = instructions/programs (system vs application software).
2. Data = raw unprocessed facts; information = processed, meaningful data.
3. Spreadsheet functions: `=SUM(range)` , `=AVERAGE(range)` , `=IF(condition, true, false)` .
4. Database: organised, structured data in tables, efficient for querying large volumes.
5. LAN = small area network; WAN = large area network (internet is the largest WAN).
6. Security measures: passwords/access control, antivirus, backups, firewalls, user awareness training.
7. AIS: people + processes + technology collecting/recording/processing/reporting financial data.
8. Good information: accurate, timely, relevant, complete, appropriately presented.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.